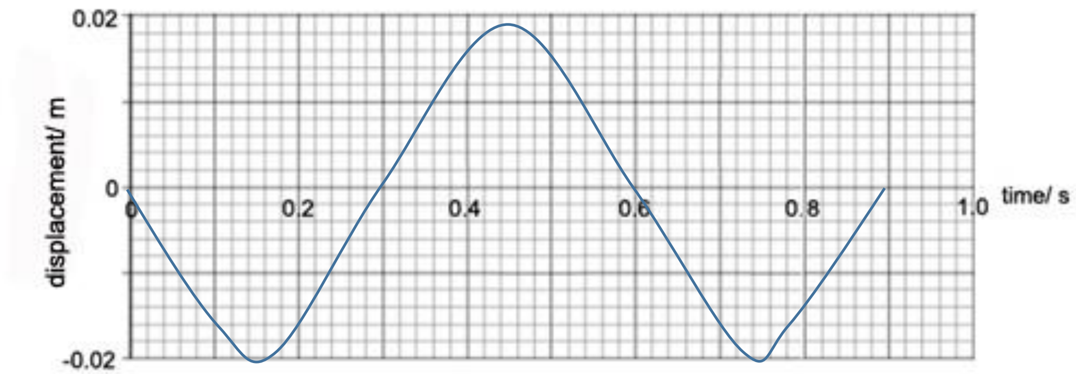
7	**(a)**	Any 2 below progressive: all particles have same amplitude stationary: amplitude varies from maximum at antinode to minimum/zero amplitude at node progressive: adjacent particles are not in phase stationary: between adjacent nodes, wave particles are in phase opposite side of a node, wave particles are anti-phase/π rad out of phase progressive: 1 wavelength is distance between two adjacent particles in phase stationary: 1wavelength is 2x distance between adjacent nodes or antinodes	B2
	(b)	Sinusoidal of correct period and amplitude 0.020 m drawn from t = 0 to t = 0.9 s	



		Workings: λ = 0.60/2.5 = 0.24 m T = 0.60 [s] correct period at 0.60 s	B1
		Correct amplitude = 0.02 m	B1
		Graph at correct phase ($-\sin \omega t$)	B1
	(c)(i)	5 ms = ¼ of period, so stationary wave at maximum displacement Note: **draw a line** – does not literally mean draw a straight line, it may be a curve/waveform	

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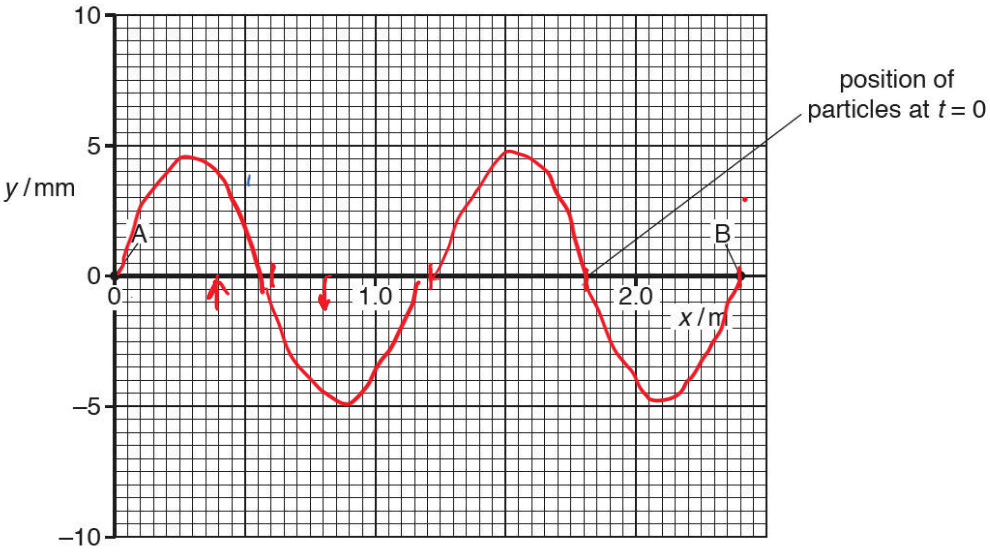


B1
B1

Correct amplitude (5.0 mm) and correct wavelength 1.2 m
Correct zero displacement/nodes at 0.0, 0.60 m, 1.2 m, 1.8 m, 2.4 m)

(c)(ii) π rad A1
(Particles on opposite side of a node are antiphase)

(c)(iii) at t = 0 s particle has maximum kinetic energy as particle is moving B1
at t = 5.0 ms no kinetic energy as particle is stationary
so decrease in kinetic energy (between t = 0 and t = 5.0 ms) B1

Comment: Do not mix up the two terms: displacement and amplitude, e.g. an antinode particle has maximum amplitude but its displacement can vary from x = 0 to $x = x_0$. It is in s.h.m.

(d)(i) Path difference between the two sources is zero or meet in phase at any point on the centre line, so waves undergo constructive difference. B1

(ii)1 Signal is from B. A1
The intensity of signal is lower as the signal travels further spreads over a larger surface area as it is further from the AGV. M1

2 phase difference = 4/24 x 2π = π/3 rad or 1.05 rad B1

3
Dist. from A to AGV = $\sqrt{7.00^2 + 50.00^2}$ = 50.49 m
C1
Dist. from B to AGV = $\sqrt{5.00^2 + 50.00^2}$ = 50.25 m

$\frac{\Delta x}{\lambda} = \frac{\Delta\phi}{2\pi} = \frac{\pi}{3(2\pi)}$ since $\Delta\phi$ = π/3.0 from graph

$$\Delta x = \frac{\lambda}{6}$$

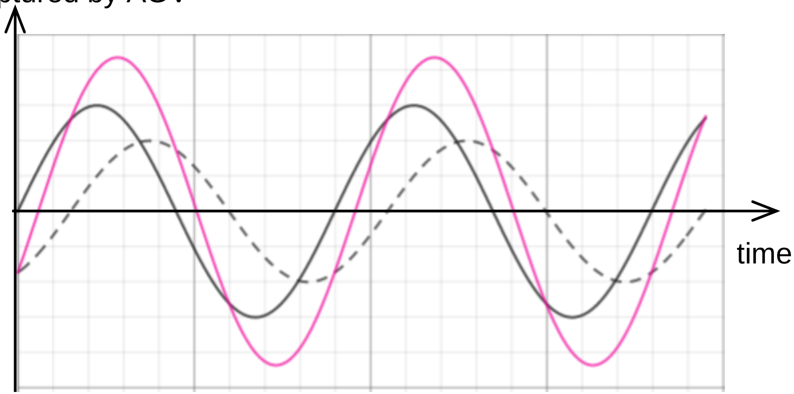
Path Difference = $\frac{\lambda}{6}$

= 50.49 – 50.25 = 0.24 m C1

λ = 1.4 m

f = c / λ = 3.0 x 10^8/ 1.4 C1

= 2.1 x 10^8 Hz A1

4		B 1- For shape B1- For accuracy of graph as given by crosses

8(a)(i) Binding energy of a nucleus is the energy released when the nucleus B1